SRI: N05LO

The spectra are evaluated at $-k_*\eta_* = 1$, through relative amplitude order $p = 5$. The Hubble-flow parameters are denoted by ϵ_j and $M_{\rm Pl}$ is the reduced Planck mass. Coefficients are generated numerically from the normalized mode equations, using the iterative algebra. The convention for the spectral derivatives is

$$\alpha_n^A = \frac{d^{n+1}\ln P_A}{d(\ln k)^{n+1}}, \qquad \alpha_0^s = n_s - 1, \qquad \alpha_0^t = n_t, \qquad r = P_t/P_s.$$

1 Spectra

1.1 Scalar spectrum

$$P_s(k_*) = \frac{H_*^2}{8\pi^2 M_{\rm Pl}^2 \epsilon_{1*}} \left[\sum_{d=0}^{5} F_{s,d} + O(\mathrm{SR}^6)\right].$$

Relative order 0.

$$F_{s,0} = 1.0$$

Relative order 1.

$$F_{s,1} = 0.72963715453852\epsilon_2 - 0.54072569092296\epsilon_1$$

Relative order 2.

$$\begin{aligned} F_{s,2} = {} & 1.5402686460338\epsilon_1^2 + 0.49988573877771\epsilon_2^2 \\ & + 1.0192767657904\epsilon_1\epsilon_2 + 0.14504832807052\epsilon_2\epsilon_3 \end{aligned}$$

Relative order 3.

$$\begin{aligned} F_{s,3} = {} & 0.20072294544592\epsilon_2^3 + 1.6057835635673\epsilon_1^3 \\ & + 4.2977813647689\epsilon_1^2\epsilon_2 + 1.959600275274\epsilon_1\epsilon_2^2 \\ & + 0.030669313835717\epsilon_2\epsilon_3^2 - 0.23520055946694\epsilon_2^2\epsilon_3 \\ & + 0.030669313835717\epsilon_2\epsilon_3\epsilon_4 - 0.65969152604436\epsilon_1\epsilon_2\epsilon_3 \end{aligned}$$

Relative order 4.

$$\begin{aligned} F_{s,4} = {} & 0.083467005785127\epsilon_2^4 + 2.9412556561294\epsilon_1^4 \\ & + 0.055787239966231\epsilon_2\epsilon_3^3 + 8.1576664436908\epsilon_1^2\epsilon_2^2 \\ & + 9.1655563981061\epsilon_1^3\epsilon_2 + 2.2830486720555\epsilon_1\epsilon_2^3 \\ & - 0.20996470057763\epsilon_2^3\epsilon_3 - 0.058660421836305\epsilon_2^2\epsilon_3^2 \\ & + 0.055787239966231\epsilon_2\epsilon_3\epsilon_4^2 + 0.16736171989869\epsilon_2\epsilon_3^2\epsilon_4 \\ & + 0.38664546255166\epsilon_1^2\epsilon_2\epsilon_3 - 0.016062214045699\epsilon_2^2\epsilon_3\epsilon_4 \\ & - 0.079171041433698\epsilon_1\epsilon_2\epsilon_3^2 - 0.91389525644738\epsilon_1\epsilon_2^2\epsilon_3 \\ & + 0.055787239966231\epsilon_2\epsilon_3\epsilon_4\epsilon_5 \\ & - 0.079171041433698\epsilon_1\epsilon_2\epsilon_3\epsilon_4 \end{aligned}$$

Relative order 5.

$$\begin{aligned} F_{s,5} = {} & 5.0146640695735\epsilon_1^5 + 0.023060510027566\epsilon_2^5 \\ & + 27.144156780677\epsilon_1^3\epsilon_2^2 + 0.018082530239889\epsilon_2\epsilon_3^4 \\ & + 18.923628528471\epsilon_1^4\epsilon_2 + 2.4064328576811\epsilon_1\epsilon_2^4 \\ & + 0.15472627942479\epsilon_2^3\epsilon_3^2 + 14.228251540965\epsilon_1^2\epsilon_2^3 \\ & - 0.13890612714825\epsilon_2^4\epsilon_3 - 0.10759865658691\epsilon_2^2\epsilon_3^3 \end{aligned}$$

$$
\begin{aligned}
&+ 2.0842227586428\epsilon_1^2\epsilon_2\epsilon_3^2 + 0.59532818615497\epsilon_1\epsilon_2^2\epsilon_3^2 \\
&+ 0.079924208372801\epsilon_1\epsilon_2\epsilon_3^3 + 2.9656408144517\epsilon_1^3\epsilon_2\epsilon_3 \\
&+ 0.12657771167922\epsilon_2\epsilon_3^2\epsilon_4^2 + 0.06391567355745\epsilon_2^3\epsilon_3\epsilon_4 \\
&+ 0.018082530239889\epsilon_2\epsilon_3\epsilon_4^3 + 0.78872771949954\epsilon_1\epsilon_2^3\epsilon_3 \\
&+ 3.8758981992336\epsilon_1^2\epsilon_2^2\epsilon_3 + 0.0049502975692885\epsilon_2^2\epsilon_3\epsilon_4^2 \\
&+ 0.10849518143933\epsilon_2\epsilon_3^3\epsilon_4 - 0.097698061448331\epsilon_2^2\epsilon_3^2\epsilon_4 \\
&+ 2.0842227586428\epsilon_1^2\epsilon_2\epsilon_3\epsilon_4 \\
&+ 0.079924208372801\epsilon_1\epsilon_2\epsilon_3\epsilon_4^2 \\
&+ 0.018082530239889\epsilon_2\epsilon_3\epsilon_4\epsilon_5^2 \\
&+ 0.072330120959556\epsilon_2\epsilon_3^2\epsilon_4\epsilon_5 \\
&+ 1.1742909471328\epsilon_1\epsilon_2^2\epsilon_3\epsilon_4 \\
&+ 0.0049502975692885\epsilon_2^2\epsilon_3\epsilon_4\epsilon_5 \\
&+ 0.054247590719667\epsilon_2\epsilon_3\epsilon_4^2\epsilon_5 \\
&+ 0.2397726251184\epsilon_1\epsilon_2\epsilon_3^2\epsilon_4 \\
&+ 0.079924208372801\epsilon_1\epsilon_2\epsilon_3\epsilon_4\epsilon_5 \\
&+ 0.018082530239889\epsilon_2\epsilon_3\epsilon_4\epsilon_5\epsilon_6
\end{aligned}
$$

1.2 Tensor spectrum

$$
P_t(k_*) = \frac{2H_*^2}{\pi^2 M_{\rm Pl}^2}\left[\sum_{d=0}^{5} F_{t,d} + O(\mathrm{SR}^6)\right].
$$

Relative order 0.

$$
F_{t,0} = 1.0
$$

Relative order 1.

$$
F_{t,1} = -0.54072569092296\epsilon_1
$$

Relative order 2.

$$
F_{t,2} = 1.5402686460338\epsilon_1^2 - 0.25062903478191\epsilon_1\epsilon_2
$$

Relative order 3.

$$
\begin{aligned}
F_{t,3} = {} & 1.6057835635673\epsilon_1^3 + 1.8891060194179\epsilon_1^2\epsilon_2 \\
& - 0.18929040711048\epsilon_1\epsilon_2^2 - 0.18929040711048\epsilon_1\epsilon_2\epsilon_3
\end{aligned}
$$

Relative order 4.

$$
\begin{aligned}
F_{t,4} = {} & 2.9412556561294\epsilon_1^4 + 4.0859368676311\epsilon_1^3\epsilon_2 \\
& + 1.8087367742501\epsilon_1^2\epsilon_2^2 - 0.077715927178017\epsilon_1\epsilon_2^3 \\
& + 1.696905383796\epsilon_1^2\epsilon_2\epsilon_3 - 0.077715927178017\epsilon_1\epsilon_2\epsilon_3^2 \\
& - 0.23314778153405\epsilon_1\epsilon_2^2\epsilon_3 - 0.077715927178017\epsilon_1\epsilon_2\epsilon_3\epsilon_4
\end{aligned}
$$

Relative order 5.

$$
\begin{aligned}
F_{t,5} = {} & 5.0146640695735\epsilon_1^5 + 10.182162879741\epsilon_1^3\epsilon_2^2 \\
& + 1.9912795134148\epsilon_1^2\epsilon_2^3 + 9.3282240106668\epsilon_1^4\epsilon_2 \\
& - 0.041550866698239\epsilon_1\epsilon_2^4 + 5.715558726935\epsilon_1^2\epsilon_2^2\epsilon_3
\end{aligned}
$$

$$
\begin{aligned}
&+ 6.2270085551926\epsilon_1^3\epsilon_2\epsilon_3 + 1.8621396067601\epsilon_1^2\epsilon_2\epsilon_3^2 \\
&- 0.041550866698239\epsilon_1\epsilon_2\epsilon_3^3 - 0.29085606688767\epsilon_1\epsilon_2^2\epsilon_3^2 \\
&- 0.24930520018943\epsilon_1\epsilon_2^3\epsilon_3 + 1.8621396067601\epsilon_1^2\epsilon_2\epsilon_3\epsilon_4 \\
&- 0.041550866698239\epsilon_1\epsilon_2\epsilon_3\epsilon_4^2 \\
&- 0.16620346679296\epsilon_1\epsilon_2^2\epsilon_3\epsilon_4 \\
&- 0.12465260009472\epsilon_1\epsilon_2\epsilon_3^2\epsilon_4 \\
&- 0.041550866698239\epsilon_1\epsilon_2\epsilon_3\epsilon_4\epsilon_5
\end{aligned}
$$

2 Consistency relations

2.1 $n = 0$

For $p = 5$, the absolute cutoff is $W = 6$:

$$\alpha_0^t = \sum_{d=1}^{6} R_{0,d} + O(\mathrm{SR}^7).$$

Absolute order 1.

$$R_{0,1} = -0.125r$$

Absolute order 2.

$$R_{0,2} = 0.0078125r^2 + 0.125\alpha_0^s r$$

Absolute order 3.

$$
\begin{aligned}
R_{0,3} = &-0.0030712246746565r^3 + 0.125\alpha_1^s r \\
&- 0.10280837917801\left(\alpha_0^s\right)^2 r - 0.041327094794504\alpha_0^s r^2
\end{aligned}
$$

Absolute order 4.

$$
\begin{aligned}
R_{0,4} = &\, 0.0014926374043055r^4 + 0.125\alpha_2^s r \\
&+ 0.065244100954277\left(\alpha_0^s\right)^3 r + 0.024554893470954\alpha_0^s r^3 \\
&+ 0.10320649151137\left(\alpha_0^s\right)^2 r^2 - 0.060109233906372\alpha_1^s r^2 \\
&- 0.23087387125098\alpha_0^s\alpha_1^s r
\end{aligned}
$$

Absolute order 5.

$$
\begin{aligned}
R_{0,5} = &-0.00093280100983415r^5 + 0.125\alpha_3^s r \\
&+ 0.043159513090910\alpha_1^s r^3 - 0.038536574146577\left(\alpha_0^s\right)^4 r \\
&- 0.018206214478227\alpha_0^s r^4 - 0.077020534433109\alpha_2^s r^2 \\
&- 0.14103704832205\left(\alpha_1^s\right)^2 r - 0.18695785481671\left(\alpha_0^s\right)^3 r^2 \\
&- 0.10777687087114\left(\alpha_0^s\right)^2 r^3 + 0.33322659757276\alpha_0^s\alpha_1^s r^2 \\
&+ 0.25691700713838\left(\alpha_0^s\right)^2\alpha_1^s r - 0.24116427546487\alpha_0^s\alpha_2^s r
\end{aligned}
$$

Absolute order 6.

$$
\begin{aligned}
R_{0,6} = &\, 0.00070298503785145r^6 + 0.125\alpha_4^s r \\
&+ 0.020902477635746\left(\alpha_0^s\right)^5 r + 0.015994052232973\alpha_0^s r^5 \\
&+ 0.065362321729807\alpha_2^s r^3 + 0.28417656846613\left(\alpha_0^s\right)^4 r^2 \\
&+ 0.30605471800346\left(\alpha_1^s\right)^2 r^2 + 0.34969699644176\left(\alpha_0^s\right)^3 r^3
\end{aligned}
$$

$$
\begin{aligned}
&+ 0.12218812205202\left(\alpha_0^s\right)^2 r^4 - 0.093222530607224\alpha_3^s r^2 \\
&- 0.037217907105313\alpha_1^s r^4 + 0.4688109212146\alpha_0^s\alpha_2^s r^2 \\
&+ 0.37293859690816\alpha_0^s\left(\alpha_1^s\right)^2 r + 0.29046664127673\left(\alpha_0^s\right)^2\alpha_2^s r \\
&- 0.22801627965519\left(\alpha_0^s\right)^3\alpha_1^s r - 1.015918735308\left(\alpha_0^s\right)^2\alpha_1^s r^2 \\
&- 0.31399132156735\alpha_1^s\alpha_2^s r - 0.42145582743858\alpha_0^s\alpha_1^s r^3 \\
&- 0.24578024485779\alpha_0^s\alpha_3^s r
\end{aligned}
$$

2.2 $n = 1$

For $p = 5$, the absolute cutoff is $W = 7$:

$$\alpha_1^t = \sum_{d=2}^{7} R_{1,d} + O(\mathrm{SR}^8).$$

Absolute order 2.

$$R_{1,2} = 0.015625r^2 + 0.125\alpha_0^s r$$

Absolute order 3.

$$
\begin{aligned}
R_{1,3} = &-0.0029296875r^3 + 0.125\alpha_1^s r \\
&- 0.125\left(\alpha_0^s\right)^2 r - 0.046875\alpha_0^s r^2
\end{aligned}
$$

Absolute order 4.

$$
\begin{aligned}
R_{1,4} = &\, 0.0016576826498282r^4 + 0.125\alpha_2^s r \\
&+ 0.12398128438351\left(\alpha_0^s\right)^2 r^2 + 0.10280837917801\left(\alpha_0^s\right)^3 r \\
&+ 0.027641022071908\alpha_0^s r^3 - 0.072577094794504\alpha_1^s r^2 \\
&- 0.33061675835603\alpha_0^s\alpha_1^s r
\end{aligned}
$$

Absolute order 5.

$$
\begin{aligned}
R_{1,5} = &-0.0010528680915447r^5 + 0.125\alpha_3^s r \\
&+ 0.050025543685844\alpha_1^s r^3 - 0.091359233906372\alpha_2^s r^2 \\
&- 0.065244100954277\left(\alpha_0^s\right)^4 r - 0.13027434666455\left(\alpha_0^s\right)^2 r^3 \\
&- 0.021075080402355\alpha_0^s r^4 - 0.23087387125098\left(\alpha_1^s\right)^2 r \\
&- 0.24842610305581\left(\alpha_0^s\right)^3 r^2 + 0.42660617411381\left(\alpha_0^s\right)^2\alpha_1^s r \\
&+ 0.41559991864823\alpha_0^s\alpha_1^s r^2 - 0.35587387125098\alpha_0^s\alpha_2^s r
\end{aligned}
$$

Absolute order 6.

$$
\begin{aligned}
R_{1,6} = &\, 0.00079786539870909r^6 + 0.125\alpha_4^s r \\
&+ 0.038536574146577\left(\alpha_0^s\right)^5 r + 0.41043044123784\left(\alpha_0^s\right)^4 r^2 \\
&+ 0.14689479099556\left(\alpha_0^s\right)^2 r^4 + 0.018569586968286\alpha_0^s r^5 \\
&+ 0.074971901003326\alpha_2^s r^3 + 0.38411085965327\left(\alpha_1^s\right)^2 r^2 \\
&+ 0.44641771599844\left(\alpha_0^s\right)^3 r^3 - 0.043199996920584\alpha_1^s r^4 \\
&- 0.10827053443311\alpha_3^s r^2 + 0.5788087353052\alpha_0^s\alpha_2^s r^2 \\
&+ 0.65487106259881\alpha_0^s\left(\alpha_1^s\right)^2 r + 0.49808128260325\left(\alpha_0^s\right)^2\alpha_2^s r \\
&- 0.36616427546487\alpha_0^s\alpha_3^s r - 0.52323837210896\alpha_1^s\alpha_2^s r \\
&- 0.51344198472507\alpha_0^s\alpha_1^s r^3 - 1.3749765739875\left(\alpha_0^s\right)^2\alpha_1^s r^2 \\
&- 0.41106330372469\left(\alpha_0^s\right)^3\alpha_1^s r
\end{aligned}
$$

Absolute order 7.

$$\begin{aligned}R_{1,7} = &-0.00069821403713025r^7 + 0.125\alpha_5^s r \\ &+ 0.37293859690816\left(\alpha_1^s\right)^3 r + 0.042797105843338\alpha_1^s r^5 \\ &+ 0.10325045820752\alpha_3^s r^3 - 0.75129206502539\left(\alpha_0^s\right)^3 r^4 \\ &- 0.12447253060722\alpha_4^s r^2 - 0.17935279734314\left(\alpha_0^s\right)^2 r^5 \\ &- 0.020902477635746\left(\alpha_0^s\right)^6 r - 0.56207286522481\left(\alpha_1^s\right)^2 r^3 \\ &- 0.073841572008579\alpha_2^s r^4 - 0.018589066236159\alpha_0^s r^6 \\ &- 0.59662818041792\left(\alpha_0^s\right)^5 r^2 - 0.31399132156735\left(\alpha_2^s\right)^2 r \\ &- 1.2665901227865\left(\alpha_0^s\right)^4 r^3 + 1.1906681876134\alpha_1^s\alpha_2^s r^2 \\ &+ 0.53624688613452\left(\alpha_0^s\right)^2\alpha_3^s r + 0.7479510436435\alpha_0^s\alpha_3^s r^2 \\ &+ 0.66101914285082\alpha_0^s\alpha_1^s r^4 + 0.33252866783392\left(\alpha_0^s\right)^4\alpha_1^s r \\ &+ 2.9112751359658\left(\alpha_0^s\right)^2\alpha_1^s r^3 + 3.3535596284132\left(\alpha_0^s\right)^3\alpha_1^s r^2 \\ &- 2.1121504017171\left(\alpha_0^s\right)^2\alpha_2^s r^2 - 0.84337953675004\alpha_0^s\alpha_2^s r^3 \\ &- 0.51848292093192\left(\alpha_0^s\right)^3\alpha_2^s r \\ &- 1.0569874358737\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r - 0.55977156642514\alpha_1^s\alpha_3^s r \\ &- 2.8301592857432\alpha_0^s\left(\alpha_1^s\right)^2 r^2 - 0.37078024485779\alpha_0^s\alpha_4^s r \\ &+ 1.6408017979371\alpha_0^s\alpha_1^s\alpha_2^s r\end{aligned}$$

2.3 $n = 2$

For $p = 5$, the absolute cutoff is $W = 8$:

$$\alpha_2^t = \sum_{d=3}^{8} R_{2,d} + O(\mathrm{SR}^9).$$

Absolute order 3.

$$\begin{aligned}R_{2,3} = &-0.00390625r^3 + 0.125\alpha_1^s r \\ &- 0.125\left(\alpha_0^s\right)^2 r - 0.046875\alpha_0^s r^2\end{aligned}$$

Absolute order 4.

$$\begin{aligned}R_{2,4} = &\ 0.0013427734375r^4 + 0.125\alpha_2^s r \\ &+ 0.125\left(\alpha_0^s\right)^3 r + 0.125\left(\alpha_0^s\right)^2 r^2 \\ &+ 0.025390625\alpha_0^s r^3 - 0.0625\alpha_1^s r^2 \\ &- 0.375\alpha_0^s\alpha_1^s r\end{aligned}$$

Absolute order 5.

$$\begin{aligned}R_{2,5} = &-0.00099348164677838r^5 + 0.125\alpha_3^s r \\ &+ 0.050668108270534\alpha_1^s r^3 - 0.13499234851023\left(\alpha_0^s\right)^2 r^3 \\ &- 0.28928966356152\left(\alpha_0^s\right)^3 r^2 - 0.020502543360439\alpha_0^s r^4 \\ &- 0.10280837917801\left(\alpha_0^s\right)^4 r - 0.33061675835603\left(\alpha_1^s\right)^2 r \\ &- 0.088202094794504\alpha_2^s r^2 + 0.63904189589007\left(\alpha_0^s\right)^2\alpha_1^s r \\ &+ 0.46569385315053\alpha_0^s\alpha_1^s r^2 - 0.45561675835603\alpha_0^s\alpha_2^s r\end{aligned}$$

Absolute order 6.

$$
\begin{aligned}
R_{2,6} = {} & 0.0007834832445242 r^6 + 0.125 \alpha_4^s r \\
& + 0.065244100954277 \left(\alpha_0^s\right)^5 r + 0.51447193930681 \left(\alpha_0^s\right)^3 r^3 \\
& + 0.018883632267472 \alpha_0^s r^5 + 0.077748164662437 \alpha_2^s r^3 \\
& + 0.5388653261447 \left(\alpha_0^s\right)^4 r^2 + 0.15691226448912 \left(\alpha_0^s\right)^2 r^4 \\
& + 0.4600841525546 \left(\alpha_1^s\right)^2 r^2 - 0.10698423390637 \alpha_3^s r^2 \\
& - 0.044329625847117 \alpha_1^s r^4 + 0.78248004536479 \left(\alpha_0^s\right)^2 \alpha_2^s r \\
& + 1.0840862194786 \alpha_0^s \left(\alpha_1^s\right)^2 r + 0.67405262036735 \alpha_0^s \alpha_2^s r^2 \\
& - 0.56686562073618 \alpha_0^s \alpha_1^s r^3 - 0.68758257793092 \left(\alpha_0^s\right)^3 \alpha_1^s r \\
& - 0.81762161375293 \alpha_1^s \alpha_2^s r - 1.7284662943263 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \\
& - 0.48087387125098 \alpha_0^s \alpha_3^s r
\end{aligned}
$$

Absolute order 7.

$$
\begin{aligned}
R_{2,7} = {} & -0.00070216008727898 r^7 + 0.125 \alpha_5^s r \\
& + 0.1069223471116 \alpha_3^s r^3 + 0.65487106259881 \left(\alpha_1^s\right)^3 r \\
& + 0.044680319835584 \alpha_1^s r^5 - 0.038536574146577 \left(\alpha_0^s\right)^6 r \\
& - 0.076631375424485 \alpha_2^s r^4 - 0.85737561408011 \left(\alpha_0^s\right)^5 r^2 \\
& - 0.52323837210896 \left(\alpha_2^s\right)^2 r - 0.019279820845041 \alpha_0^s r^6 \\
& - 1.5778098507287 \left(\alpha_0^s\right)^4 r^3 - 0.12389553443311 \alpha_4^s r^2 \\
& - 0.64133873745452 \left(\alpha_1^s\right)^2 r^3 - 0.193795881651 \left(\alpha_0^s\right)^2 r^5 \\
& - 0.85944179815707 \left(\alpha_0^s\right)^3 r^4 + 1.4436852511254 \alpha_1^s \alpha_2^s r^2 \\
& + 4.6123541905271 \left(\alpha_0^s\right)^3 \alpha_1^s r^2 + 0.72830526168963 \alpha_0^s \alpha_1^s r^4 \\
& + 0.87237033860452 \alpha_0^s \alpha_3^s r^2 + 0.60374617445758 \left(\alpha_0^s\right)^4 \alpha_1^s r \\
& + 0.86424555806812 \left(\alpha_0^s\right)^2 \alpha_3^s r + 3.481878776513 \left(\alpha_0^s\right)^2 \alpha_1^s r^3 \\
& - 0.88940264757383 \alpha_1^s \alpha_3^s r - 0.94272853191881 \alpha_0^s \alpha_2^s r^3 \\
& - 1.8880609737729 \left(\alpha_0^s\right)^2 \left(\alpha_1^s\right)^2 r - 0.49116427546487 \alpha_0^s \alpha_4^s r \\
& - 2.69796002066 \left(\alpha_0^s\right)^2 \alpha_2^s r^2 - 0.90914458632794 \left(\alpha_0^s\right)^3 \alpha_2^s r \\
& - 3.7167089437539 \alpha_0^s \left(\alpha_1^s\right)^2 r^2 + 2.8291430625131 \alpha_0^s \alpha_1^s \alpha_2^s r
\end{aligned}
$$

Absolute order 8.

$$
\begin{aligned}
R_{2,8} = {} & 0.00070456920068177 r^8 + 0.125 \alpha_6^s r \\
& + 0.13925140335932 \alpha_4^s r^3 + 1.2215314043215 \left(\alpha_0^s\right)^6 r^2 \\
& + 0.020902477635746 \left(\alpha_0^s\right)^7 r + 1.4381221889498 \left(\alpha_0^s\right)^3 r^5 \\
& + 0.92219068354551 \left(\alpha_1^s\right)^2 r^4 + 3.8323296829683 \left(\alpha_0^s\right)^4 r^4 \\
& + 0.25673070632081 \left(\alpha_0^s\right)^2 r^6 + 4.1955594060503 \left(\alpha_0^s\right)^5 r^3 \\
& + 1.2455421028093 \left(\alpha_2^s\right)^2 r^2 + 0.085771147089679 \alpha_2^s r^5 \\
& + 0.021751280469382 \alpha_0^s r^7 - 0.11864796091522 \alpha_3^s r^4 \\
& - 0.14009753060722 \alpha_5^s r^2 - 2.9232654753033 \left(\alpha_1^s\right)^3 r^2 \\
& - 0.050080440992255 \alpha_1^s r^6 + 2.03984067706 \alpha_1^s \alpha_3^s r^2 \\
& + 2.3871021072094 \left(\alpha_0^s\right)^3 \left(\alpha_1^s\right)^2 r \\
& + 1.9547931195045 \alpha_0^s \left(\alpha_2^s\right)^2 r + 0.90702713099232 \left(\alpha_0^s\right)^2 \alpha_4^s r \\
& + 16.184733661745 \left(\alpha_0^s\right)^2 \left(\alpha_1^s\right)^2 r^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 8.5890660587772\alpha_0^s \left(\alpha_1^s\right)^2 r^3 + 6.2953483110811 \left(\alpha_0^s\right)^2 \alpha_2^s r^3 \\
&+ 0.85101158876584 \left(\alpha_0^s\right)^4 \alpha_2^s r + 1.0744936354652\alpha_0^s\alpha_4^s r^2 \\
&+ 1.3640509580442\alpha_0^s\alpha_2^s r^4 + 2.7596175886616 \left(\alpha_1^s\right)^2 \alpha_2^s r \\
&+ 7.8289781977937 \left(\alpha_0^s\right)^3 \alpha_2^s r^2 - 0.99377356769359\alpha_0^s\alpha_1^s r^5 \\
&- 2.4869134686556\alpha_0^s \left(\alpha_1^s\right)^3 r - 1.1877542095598\alpha_2^s\alpha_3^s r \\
&- 1.4062650501208\alpha_0^s\alpha_3^s r^3 - 15.480967047284 \left(\alpha_0^s\right)^3 \alpha_1^s r^3 \\
&- 9.9459636515628 \left(\alpha_0^s\right)^4 \alpha_1^s r^2 - 3.7800524622085 \left(\alpha_0^s\right)^2 \alpha_3^s r^2 \\
&- 1.0547298070664 \left(\alpha_0^s\right)^3 \alpha_3^s r - 2.3372176459017\alpha_1^s\alpha_2^s r^3 \\
&- 6.4975922521103 \left(\alpha_0^s\right)^2 \alpha_1^s r^4 - 0.4579435336484 \left(\alpha_0^s\right)^5 \alpha_1^s r \\
&- 0.93055181128293\alpha_1^s\alpha_4^s r - 0.49578024485779\alpha_0^s\alpha_5^s r \\
&+ 3.2730671366313\alpha_0^s\alpha_1^s\alpha_3^s r - 5.3102254324804 \left(\alpha_0^s\right)^2 \alpha_1^s\alpha_2^s r \\
&- 12.720525783639\alpha_0^s\alpha_1^s\alpha_2^s r^2
\end{aligned}
$$

2.4 $n = 3$

For $p = 5$, the absolute cutoff is $W = 9$:

$$\alpha_3^t = \sum_{d=4}^{9} R_{3,d} + O(\mathrm{SR}^{10}).$$

Absolute order 4.

$$
\begin{aligned}
R_{3,4} = &0.00146484375r^4 + 0.125\alpha_2^s r \\
&+ 0.125 \left(\alpha_0^s\right)^3 r + 0.0234375\alpha_0^s r^3 \\
&+ 0.109375 \left(\alpha_0^s\right)^2 r^2 - 0.0625\alpha_1^s r^2 \\
&- 0.375\alpha_0^s\alpha_1^s r
\end{aligned}
$$

Absolute order 5.

$$
\begin{aligned}
R_{3,5} = &-0.000762939453125r^5 + 0.125\alpha_3^s r \\
&+ 0.0419921875\alpha_1^s r^3 - 0.125 \left(\alpha_0^s\right)^4 r \\
&- 0.375 \left(\alpha_1^s\right)^2 r - 0.078125\alpha_2^s r^2 \\
&- 0.28125 \left(\alpha_0^s\right)^3 r^2 - 0.01708984375\alpha_0^s r^4 \\
&- 0.1201171875 \left(\alpha_0^s\right)^2 r^3 + 0.75 \left(\alpha_0^s\right)^2 \alpha_1^s r \\
&+ 0.4375\alpha_0^s\alpha_1^s r^2 - 0.5\alpha_0^s\alpha_2^s r
\end{aligned}
$$

Absolute order 6.

$$
\begin{aligned}
R_{3,6} = &0.00069887861331449r^6 + 0.125\alpha_4^s r \\
&+ 0.52264594794504 \left(\alpha_1^s\right)^2 r^2 + 0.10280837917801 \left(\alpha_0^s\right)^5 r \\
&+ 0.52433019631832 \left(\alpha_0^s\right)^3 r^3 + 0.017257388611671\alpha_0^s r^5 \\
&+ 0.07369519446916\alpha_2^s r^3 + 0.1495700174229 \left(\alpha_0^s\right)^2 r^4 \\
&+ 0.61990642191755 \left(\alpha_0^s\right)^4 r^2 - 0.042328393296221\alpha_1^s r^4 \\
&- 0.1038270947945\alpha_3^s r^2 + 1.6087005501362\alpha_0^s \left(\alpha_1^s\right)^2 r \\
&+ 1.0946586542461 \left(\alpha_0^s\right)^2 \alpha_2^s r + 0.71467513753404\alpha_0^s\alpha_2^s r^2 \\
&- 0.57385180946901\alpha_0^s\alpha_1^s r^3 - 1.9544879813691 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \\
&- 1.1168502750681\alpha_1^s\alpha_2^s r - 1.0502754126021 \left(\alpha_0^s\right)^3 \alpha_1^s r \\
&- 0.58061675835603\alpha_0^s\alpha_3^s r
\end{aligned}
$$

Absolute order 7.

$$
\begin{aligned}
R_{3,7} = & -0.00066040799045703 r^7 + 0.125 \alpha_5^s r \\
& + 0.044182253542082 \alpha_1^s r^5 + 0.10547078563903 \alpha_3^s r^3 \\
& + 1.0840862194786 \left(\alpha_1^s\right)^3 r - 0.12260923390637 \alpha_4^s r^2 \\
& - 0.065244100954277 \left(\alpha_0^s\right)^6 r - 1.8011431833565 \left(\alpha_0^s\right)^4 r^3 \\
& - 1.1197437723225 \left(\alpha_0^s\right)^5 r^2 - 0.076712092161027 \alpha_2^s r^4 \\
& - 0.70760825653778 \left(\alpha_1^s\right)^2 r^3 - 0.19445546235151 \left(\alpha_0^s\right)^2 r^5 \\
& - 0.018648400200653 \alpha_0^s r^6 - 0.81762161375293 \left(\alpha_2^s\right)^2 r \\
& - 0.9077746502428 \left(\alpha_0^s\right)^3 r^4 + 0.75724388321833 \alpha_0^s \alpha_1^s r^4 \\
& + 1.2633539166158 \left(\alpha_0^s\right)^2 \alpha_3^s r + 5.8694148411763 \left(\alpha_0^s\right)^3 \alpha_1^s r^2 \\
& + 3.899202830862 \left(\alpha_0^s\right)^2 \alpha_1^s r^3 + 1.0138030827023 \left(\alpha_0^s\right)^4 \alpha_1^s r \\
& + 1.7276736271957 \alpha_1^s \alpha_2^s r^2 + 0.96375532208646 \alpha_0^s \alpha_3^s r^2 \\
& - 1.0111179362879 \alpha_0^s \alpha_2^s r^3 - 0.60587387125098 \alpha_0^s \alpha_4^s r \\
& - 3.1468339532713 \left(\alpha_0^s\right)^2 \left(\alpha_1^s\right)^2 r \\
& - 4.6296729998974 \alpha_0^s \left(\alpha_1^s\right)^2 r^2 - 3.2598096829233 \left(\alpha_0^s\right)^2 \alpha_2^s r^2 \\
& - 1.2984954850039 \alpha_1^s \alpha_3^s r - 1.4700626232957 \left(\alpha_0^s\right)^3 \alpha_2^s r \\
& + 4.5507541434397 \alpha_0^s \alpha_1^s \alpha_2^s r
\end{aligned}
$$

Absolute order 8.

$$
\begin{aligned}
R_{3,8} = & \, 0.00068532023746048 r^8 + 0.125 \alpha_6^s r \\
& + 4.3619461768873 \left(\alpha_0^s\right)^4 r^4 + 0.038536574146577 \left(\alpha_0^s\right)^7 r \\
& + 5.2029254536162 \left(\alpha_0^s\right)^5 r^3 + 0.021703788578011 \alpha_0^s r^7 \\
& + 1.545267118531 \left(\alpha_0^s\right)^3 r^5 + 1.7512659597646 \left(\alpha_0^s\right)^6 r^2 \\
& + 1.010573413351 \left(\alpha_1^s\right)^2 r^4 + 0.26453516473845 \left(\alpha_0^s\right)^2 r^6 \\
& + 0.13887279321988 \alpha_4^s r^3 + 0.087120557839057 \alpha_2^s r^5 \\
& + 1.524715047639 \left(\alpha_2^s\right)^2 r^2 - 3.8575245524135 \left(\alpha_1^s\right)^3 r^2 \\
& - 0.05080989468861 \alpha_1^s r^6 - 0.13952053443311 \alpha_5^s r^2 \\
& - 0.12024763108045 \alpha_3^s r^4 + 1.1971819419038 \alpha_0^s \alpha_4^s r^2 \\
& + 10.320494578314 \left(\alpha_0^s\right)^3 \alpha_2^s r^2 + 1.4593688359096 \alpha_0^s \alpha_2^s r^4 \\
& + 4.7937562503095 \left(\alpha_1^s\right)^2 \alpha_2^s r + 10.176804101055 \alpha_0^s \left(\alpha_1^s\right)^2 r^3 \\
& + 3.3523814346221 \alpha_0^s \left(\alpha_2^s\right)^2 r + 7.272533332723 \left(\alpha_0^s\right)^2 \alpha_2^s r^3 \\
& + 4.3030456716032 \left(\alpha_0^s\right)^3 \left(\alpha_1^s\right)^2 r + 2.4584809206766 \alpha_1^s \alpha_3^s r^2 \\
& + 1.355409833533 \left(\alpha_0^s\right)^2 \alpha_4^s r \\
& + 21.892191226939 \left(\alpha_0^s\right)^2 \left(\alpha_1^s\right)^2 r^2 \\
& + 1.5128907607855 \left(\alpha_0^s\right)^4 \alpha_2^s r - 1.7733901443961 \left(\alpha_0^s\right)^3 \alpha_3^s r \\
& - 1.0496651482817 \alpha_0^s \alpha_1^s r^5 - 4.6393166739312 \left(\alpha_0^s\right)^2 \alpha_3^s r^2 \\
& - 0.61616427546487 \alpha_0^s \alpha_5^s r - 13.841809923869 \left(\alpha_0^s\right)^4 \alpha_1^s r^2 \\
& - 4.4309930101446 \alpha_0^s \left(\alpha_1^s\right)^3 r - 18.743138308883 \left(\alpha_0^s\right)^3 \alpha_1^s r^3 \\
& - 2.6475317332077 \alpha_1^s \alpha_2^s r^3 - 1.3805669230387 \alpha_1^s \alpha_4^s r \\
& - 1.5289756803498 \alpha_0^s \alpha_3^s r^3 - 0.83496561933704 \left(\alpha_0^s\right)^5 \alpha_1^s r \\
& - 7.2513766025917 \left(\alpha_0^s\right)^2 \alpha_1^s r^4 - 1.9358793917918 \alpha_2^s \alpha_3^s r
\end{aligned}
$$

$$+ 5.4470368262232\alpha_0^s\alpha_1^s\alpha_3^s r - 16.335385878746\alpha_0^s\alpha_1^s\alpha_2^s r^2$$
$$- 9.3326987690427\left(\alpha_0^s\right)^2\alpha_1^s\alpha_2^s r$$

Absolute order 9.

$$\begin{aligned}
R_{3,9} = &-0.00077806890581001r^9 + 0.125\alpha_7^s r \\
&+ 0.06358704244356\alpha_1^s r^7 + 9.5157792367703\left(\alpha_1^s\right)^3 r^3 \\
&+ 0.15020868322051\alpha_3^s r^5 + 0.17525234851113\alpha_5^s r^3 \\
&- 2.6786068018394\left(\alpha_0^s\right)^3 r^6 - 2.4869134686556\left(\alpha_1^s\right)^4 r \\
&- 1.1877542095598\left(\alpha_3^s\right)^2 r - 0.15572253060722\alpha_6^s r^2 \\
&- 1.5189127765673\left(\alpha_1^s\right)^2 r^5 - 17.933550190782\left(\alpha_0^s\right)^5 r^4 \\
&- 0.020902477635746\left(\alpha_0^s\right)^8 r - 13.339047064301\left(\alpha_0^s\right)^6 r^3 \\
&- 9.8616620914994\left(\alpha_0^s\right)^4 r^5 - 2.4713378521286\left(\alpha_0^s\right)^7 r^2 \\
&- 2.6843690618889\left(\alpha_2^s\right)^2 r^3 - 0.17465185173482\alpha_4^s r^4 \\
&- 0.38149897445917\left(\alpha_0^s\right)^2 r^7 - 0.10897896751649\alpha_2^s r^6 \\
&- 0.02733787086542\alpha_0^s r^8 + 9.6482197902839\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^3 r \\
&+ 1.5394198894557\alpha_0^s\alpha_1^s r^6 + 2.4542491246068\alpha_0^s\alpha_3^s r^4 \\
&+ 11.804717477222\left(\alpha_0^s\right)^2\alpha_3^s r^3 + 72.21470484842\left(\alpha_0^s\right)^4\alpha_1^s r^3 \\
&+ 1.4028073758501\left(\alpha_0^s\right)^2\alpha_5^s r + 3.2619032889355\alpha_1^s\alpha_4^s r^2 \\
&+ 13.367356356375\left(\alpha_0^s\right)^2\alpha_1^s r^5 + 1.9057413958323\left(\alpha_0^s\right)^4\alpha_3^s r \\
&+ 15.732895949371\left(\alpha_0^s\right)^3\alpha_3^s r^2 + 4.1955896290722\alpha_1^s\alpha_2^s r^4 \\
&+ 4.7106441588735\alpha_2^s\alpha_3^s r^2 + 0.60426087709863\left(\alpha_0^s\right)^6\alpha_1^s r \\
&+ 38.920069106575\alpha_0^s\left(\alpha_1^s\right)^3 r^2 + 1.4322862272868\alpha_0^s\alpha_5^s r^2 \\
&+ 6.0326847252929\left(\alpha_1^s\right)^2\alpha_3^s r + 49.631939118393\left(\alpha_0^s\right)^3\alpha_1^s r^4 \\
&+ 27.575781873901\left(\alpha_0^s\right)^5\alpha_1^s r^2 + 7.4740282968277\alpha_1^s\left(\alpha_2^s\right)^2 r \\
&- 0.62078024485779\alpha_0^s\alpha_6^s r - 2.2151677365237\alpha_0^s\alpha_2^s r^5 \\
&- 4.6768197754514\left(\alpha_0^s\right)^4\left(\alpha_1^s\right)^2 r \\
&- 78.451840902511\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r^3 \\
&- 20.710877845769\alpha_0^s\left(\alpha_1^s\right)^2 r^4 - 1.4263320561407\alpha_1^s\alpha_5^s r \\
&- 2.1183060208428\alpha_2^s\alpha_4^s r - 2.1443384104247\alpha_0^s\alpha_4^s r^3 \\
&- 4.3219288525125\alpha_1^s\alpha_3^s r^3 - 73.243320022063\left(\alpha_0^s\right)^3\left(\alpha_1^s\right)^2 r^2 \\
&- 37.486758705034\left(\alpha_0^s\right)^3\alpha_2^s r^3 - 1.3089551224142\left(\alpha_0^s\right)^5\alpha_2^s r \\
&- 22.035682750882\left(\alpha_1^s\right)^2\alpha_2^s r^2 - 6.1322897063433\left(\alpha_0^s\right)^2\alpha_4^s r^2 \\
&- 26.018233363777\left(\alpha_0^s\right)^4\alpha_2^s r^2 \\
&- 7.2650185519848\left(\alpha_0^s\right)^2\left(\alpha_2^s\right)^2 r \\
&- 14.97896466591\left(\alpha_0^s\right)^2\alpha_2^s r^4 - 15.608461554937\alpha_0^s\left(\alpha_2^s\right)^2 r^2 \\
&- 1.9617569380588\left(\alpha_0^s\right)^3\alpha_4^s r + 13.488476001963\left(\alpha_0^s\right)^3\alpha_1^s\alpha_2^s r \\
&+ 40.919197164152\alpha_0^s\alpha_1^s\alpha_2^s r^3 + 6.0176732098989\alpha_0^s\alpha_1^s\alpha_4^s r \\
&+ 8.3704075852001\alpha_0^s\alpha_2^s\alpha_3^s r + 82.879921620928\left(\alpha_0^s\right)^2\alpha_1^s\alpha_2^s r^2 \\
&- 20.840808859589\alpha_0^s\left(\alpha_1^s\right)^2\alpha_2^s r \\
&- 11.747481990311\left(\alpha_0^s\right)^2\alpha_1^s\alpha_3^s r - 25.047186783622\alpha_0^s\alpha_1^s\alpha_3^s r^2
\end{aligned}$$